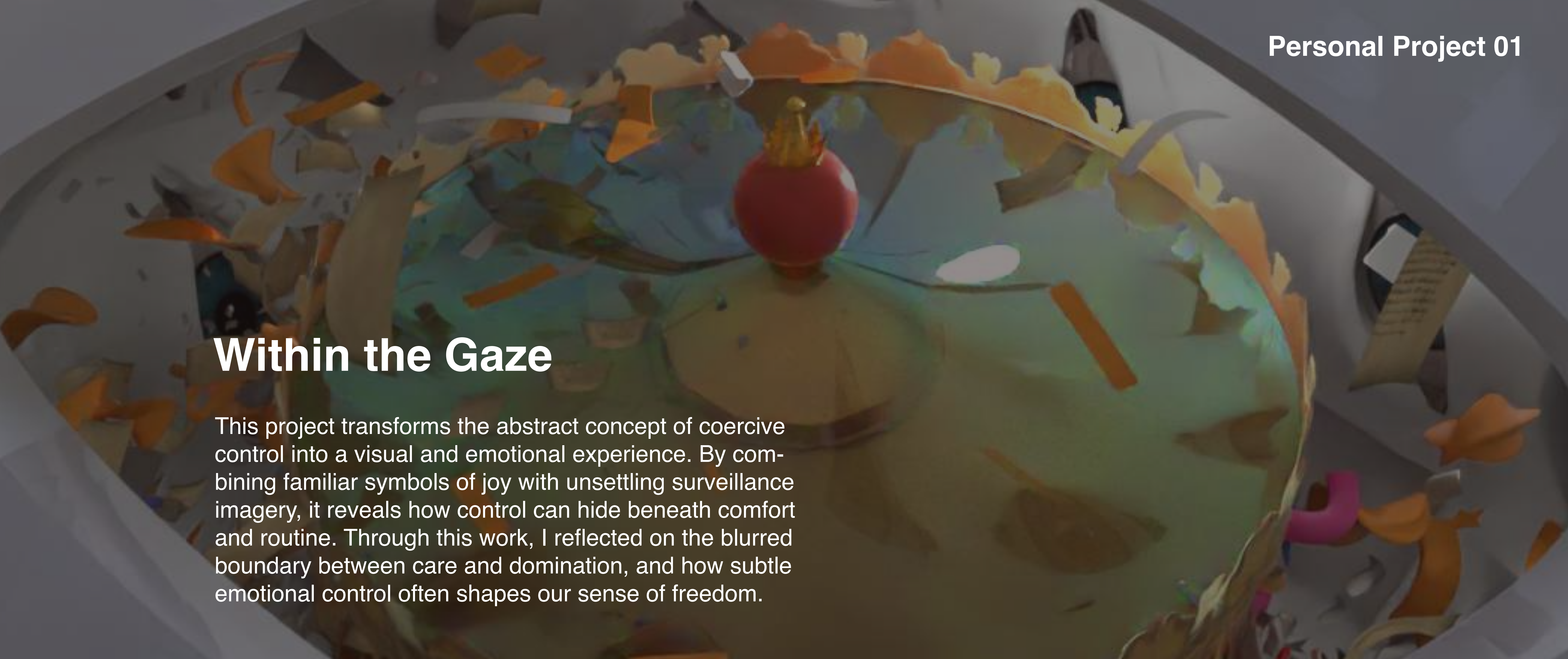
PORTFOLIO

DANNI XIE /2024–2025



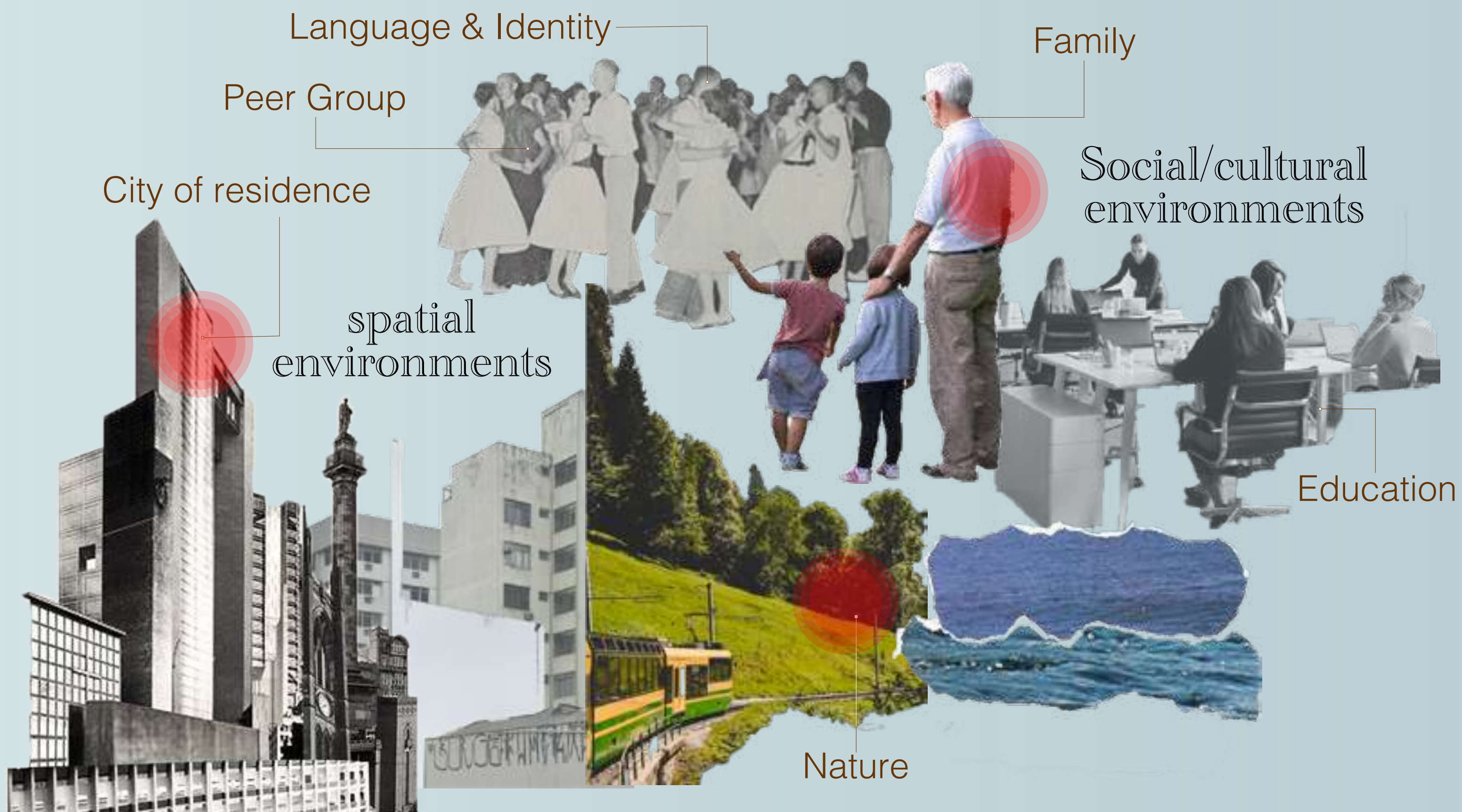
Within the Gaze

This project transforms the abstract concept of coercive control into a visual and emotional experience. By combining familiar symbols of joy with unsettling surveillance imagery, it reveals how control can hide beneath comfort and routine. Through this work, I reflected on the blurred boundary between care and domination, and how subtle emotional control often shapes our sense of freedom.

● Background

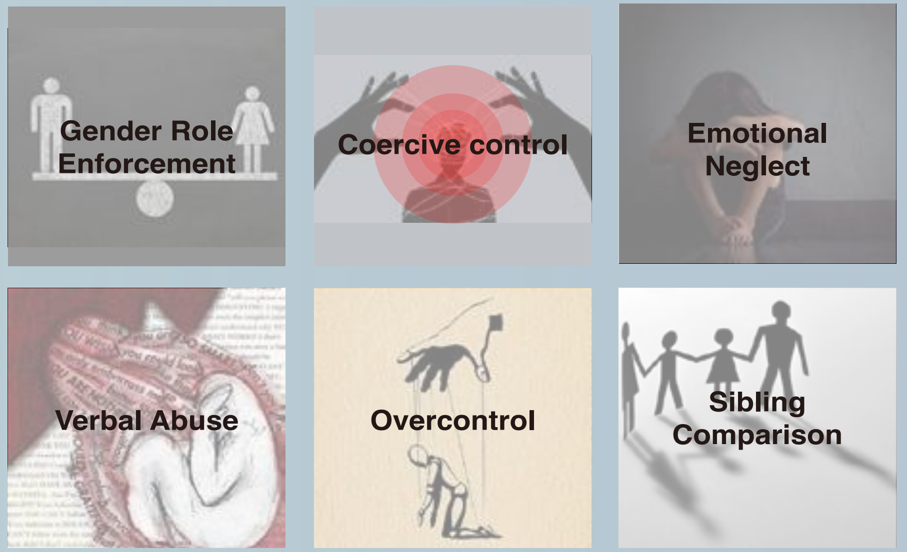
In the field of environmental psychology, Kurt Lewin (1936) famously proposed——Behavior= ***Person×Environment***. Expressed by the formula: B = f(P, E)m

How Does the Environment Shape Us?



Conclusion: Our identity is shaped by the environments we inhabit—***family, culture, and the spaces*** around us all leave subtle imprints.

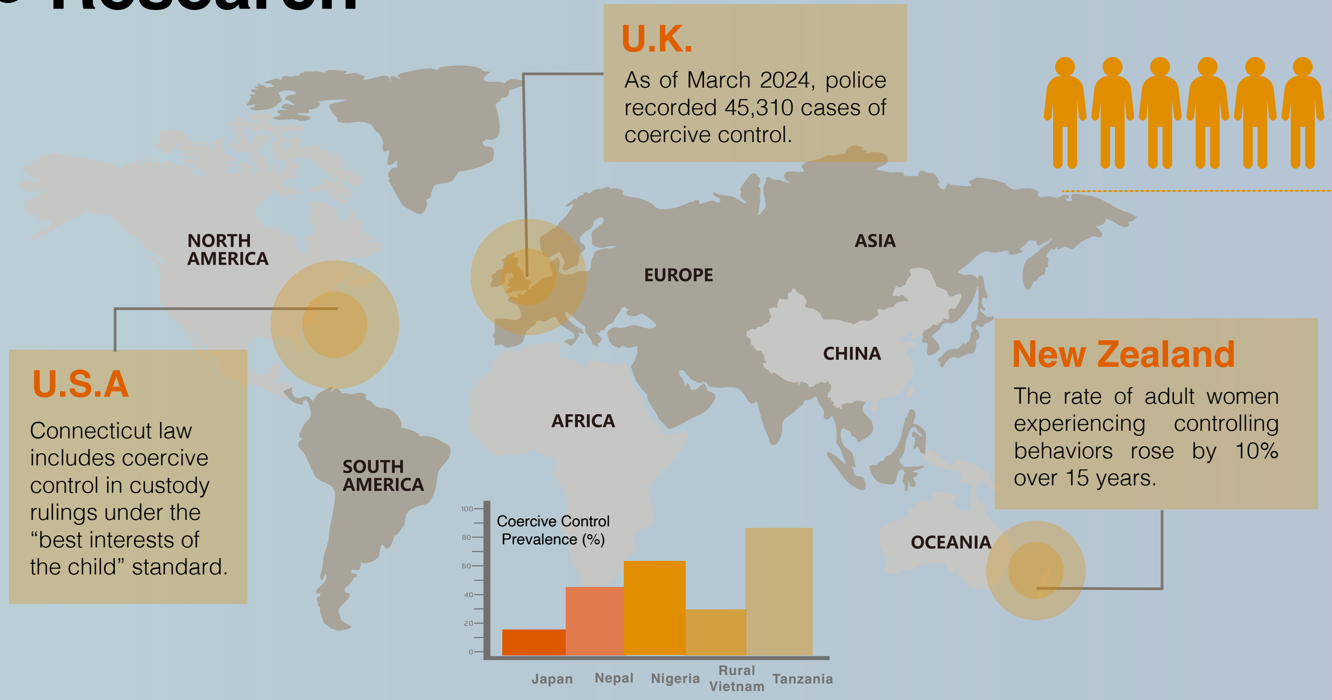
● Problems in family relationships



What Is Coercive Control?

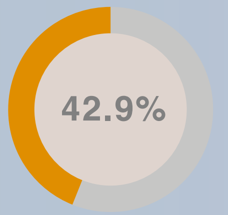
Coercive control is present in more than half of all domestic violence incidents.

● Research



Over ***60%*** of victims say coercive control is one of the hardest forms of abuse to escape.

Coercive control can cause serious psychological problems, including ***PTSD, depression, and suicidal tendencies.***



42.9% of those facing coercive control also experienced physical abuse.

Design Development

To give the audience a direct sense of the hidden oppression in familial coercive control, I have painted some ***scenes*** depicting this form of violence.

Control of Time

The Family Meal

At the Desk

Financial Control

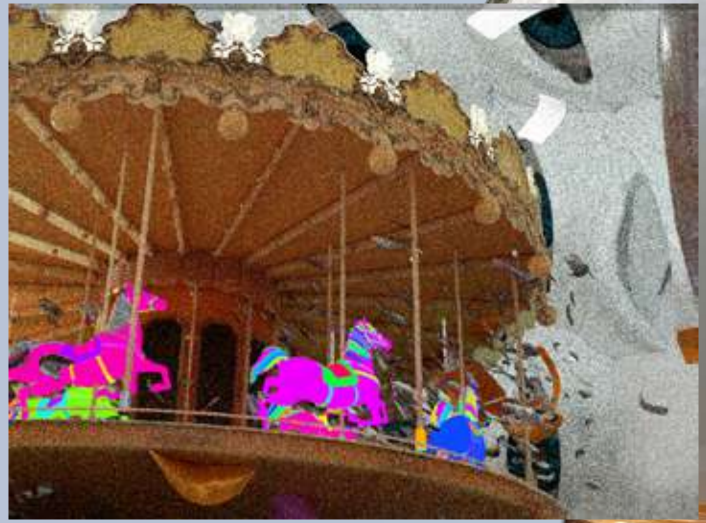
Conclusion: We can see that coercive control is a persistent form of abusive behavior that deprives a person of their sense of autonomy and independence.

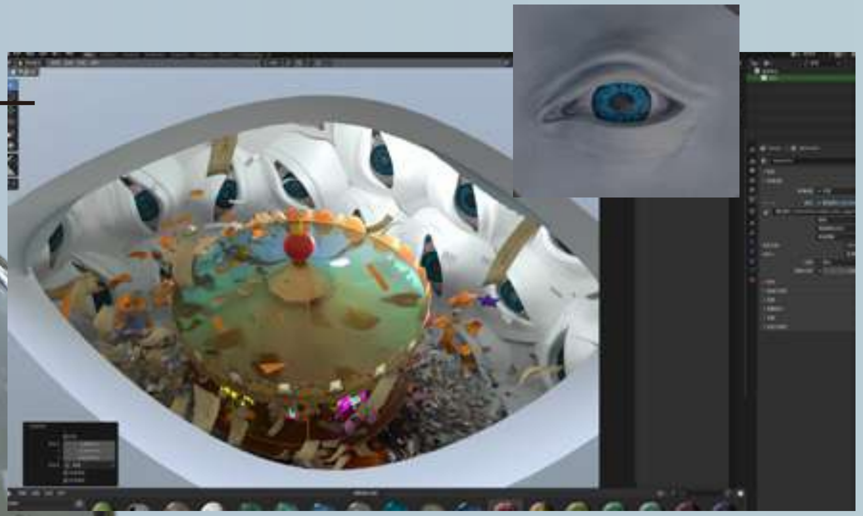
Brainstorming

Conclusion: The carrousel in the amusement park, the walk-in snow globe, and the diary—by placing these elements together within the same frame, I aim to explore an emotional contrast: can objects originally associated with tenderness, childlike wonder, and a sense of security also become shells of constraint and manipulation? For those being controlled, domination does not always appear in a serious guise. It can emerge anywhere, even hidden within scenes of joy.

Process

The crystal sphere transforms the idea of “home” into a sealed world of observation.
Its inner surface covered with watching eyes represents constant scrutiny and the pressure to perform under invisible expectations.

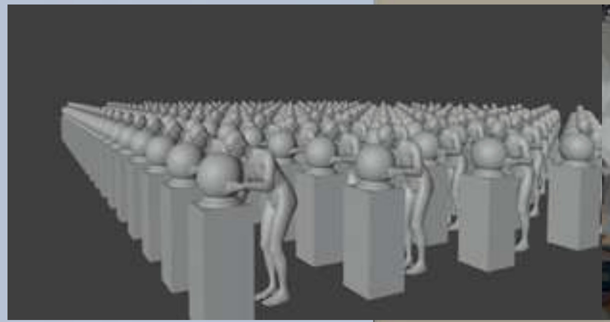


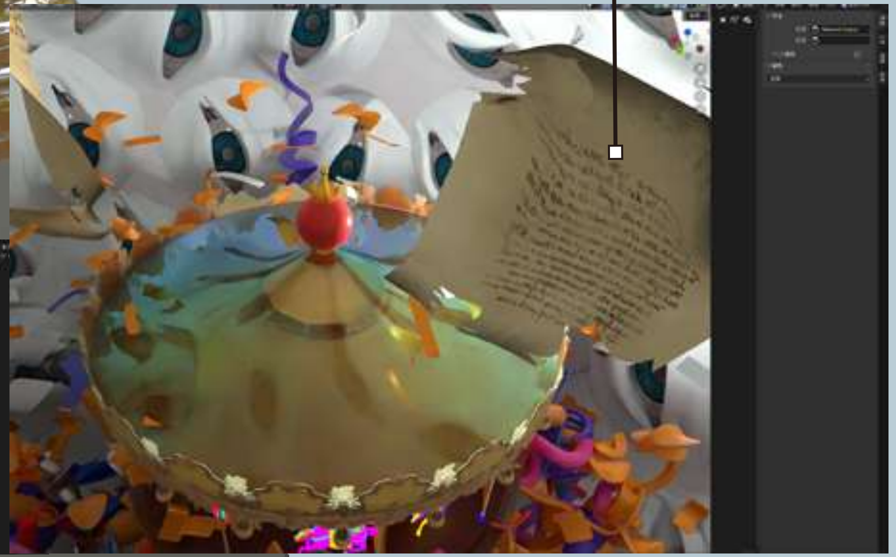


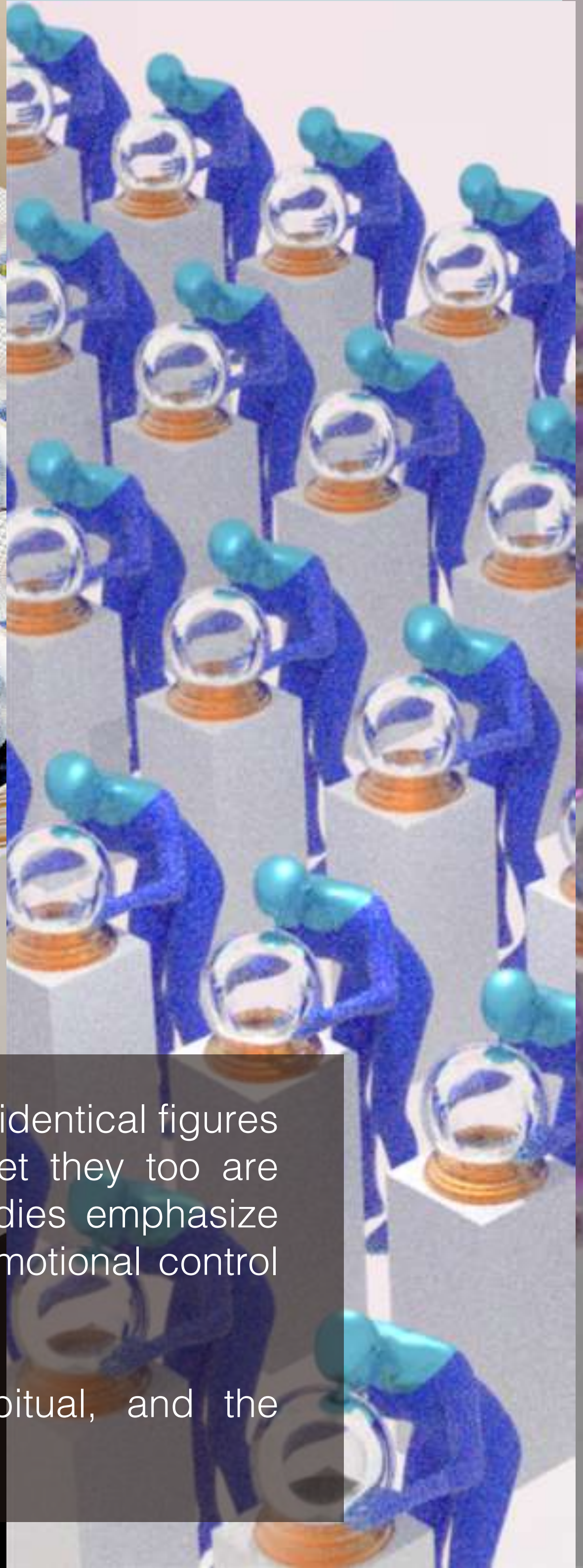
The carousel was chosen as the central metaphor for coercive control — a place of supposed joy that conceals repetition and confinement.

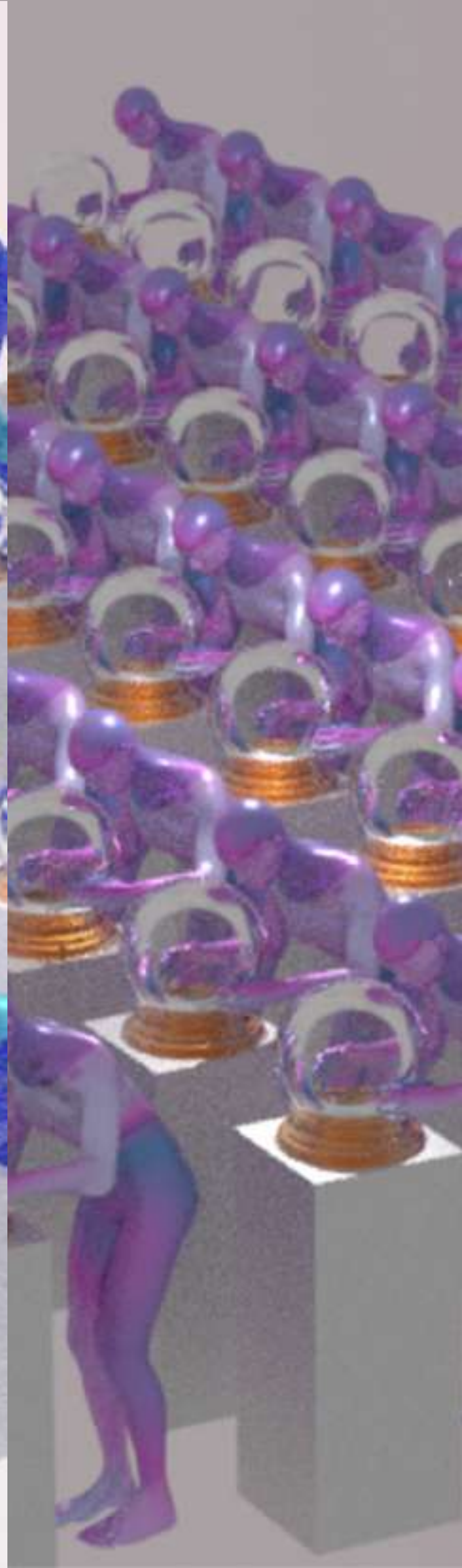
The **floating fragments** of torn diaries hint at lost privacy and the destruction of one’s inner voice.

The circular motion of carousel reflects the invisible loops of obedience and emotional manipulation that victims experience within family control.







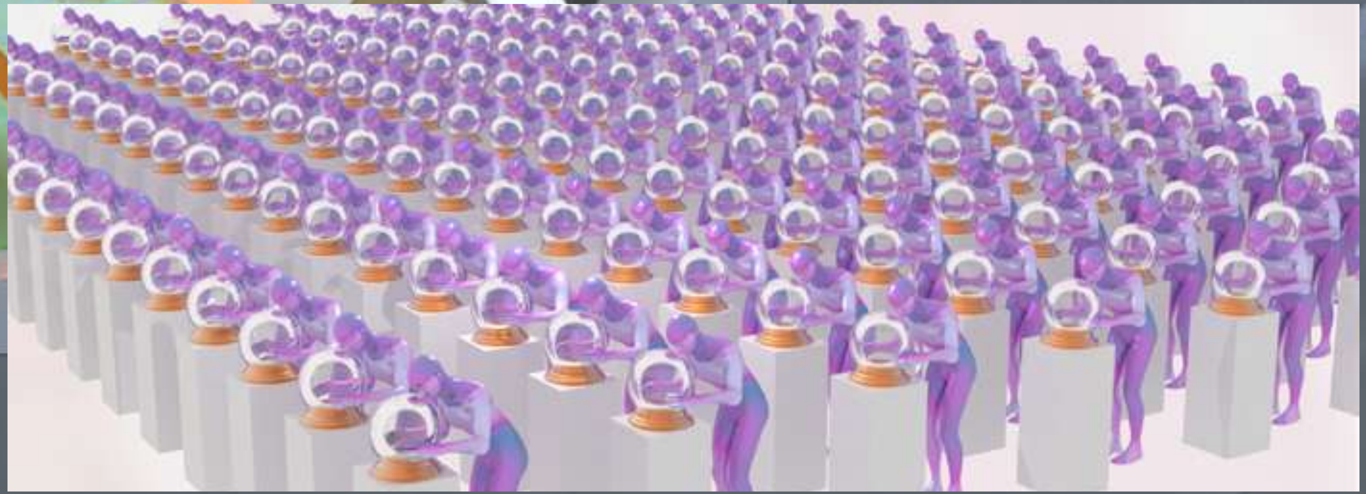


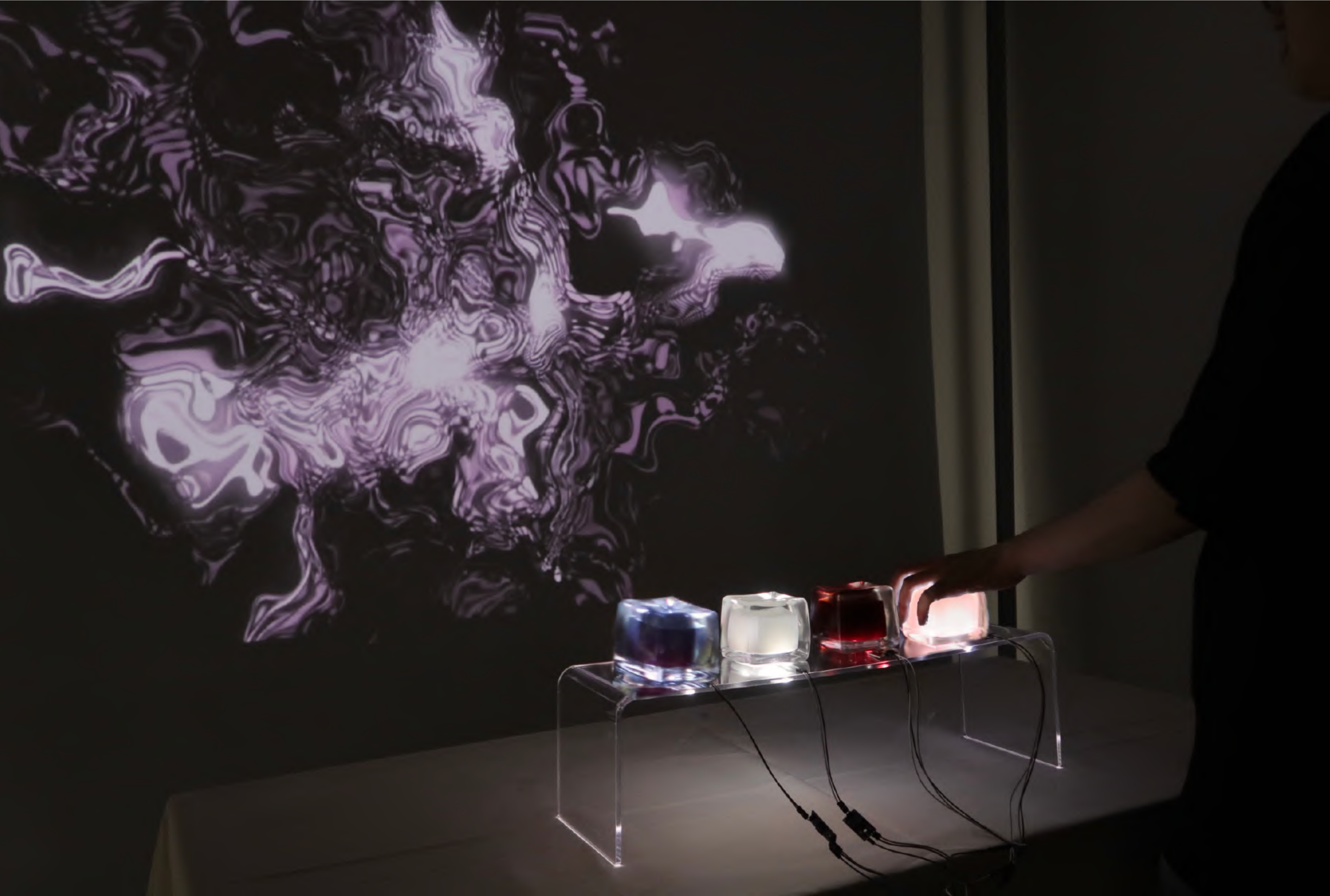
Outside the crystal sphere, identical figures bend over, watching — yet they too are watched. The mirrored bodies emphasize conformity, echoing how emotional control reproduces itself endlessly.

surveillance becomes habitual, and the controller is never truly free.

Outcome

The final work reveals a fragile balance between beauty and restraint. The carousel, the watching eyes, and the crystal sphere together form a cage nurtured within comfort.
In building this world, I realized that control is often quiet and gentle — disguised as love, yet leaving the deepest scars on the soul.





SqueezeScape

—Interactive installation

Video Link：*https://youtu.be/LOHNZ_ss11o*

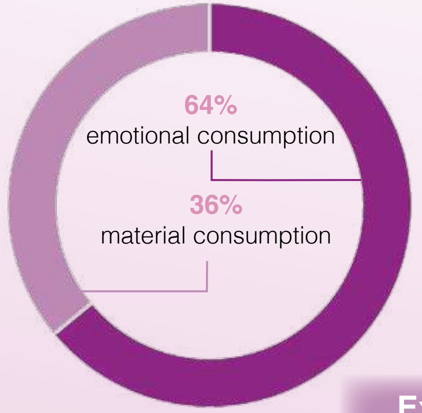
SqueezeScape is an interactive installation that explores the externalization of emotion through tactile interaction and multisensory feedback.

By squeezing soft objects embedded with pressure sensors, users trigger real-time changes in visuals, lighting, and interactive sound, each reflecting a distinct emotional state. The project invites participants to physically engage with their inner feelings, revealing the complexity and contradictions of emotional experience.

Background

The Popularity of Stress Relief Toys in the Digital Society

On Chinese social media platforms, the topic of stress relief toys has gained remarkable traction — with over 3.39 billion views on the lifestyle-sharing app REDnote and nearly 10 billion views on Douyin.



·The rise of satisfying videos reflects a shift from material consumption to emotional consumption in the digital age.

·In contrast to traditional material consumption, these digital micro-pleasures cater to our psychological needs for control, release, and comfort.

Examples...

Squishy toys

Originated from the Japanese brand *iBloom*. While the foam material has industrial roots, *iBloom*'s creative design redefined it as a form of emotional comfort and playful expression.

Slime

Slime has become a popular trend for sensory stress relief, allowing for a wide range of tactile and auditory experiences by incorporating different materials and add-ins.

Creative squishy adaptations

Squishies are also used in IP merchandise, such as museum-inspired designs that make cultural heritage more playful and accessible.

Beyond the squish

When squishing isn't enough, crushing takes over.Beyond soft squishies and slime, some find even greater satisfaction in watching objects crushed — the ultimate form of visual stress relief.

What makes viewers feel relaxed and satisfied

Personal Factors

1. **Stress Visualization:**
This type of visually satisfying content helps viewers project their stress onto the object being squeezed, enabling emotional release and regulation.

2. **Projected Sense of Control:**
In an anxious, unpredictable life, the controllable sensory experience of stress-relief toys—softness, pliability, repeatability, and predictable rebound—offers users a sense of safety and temporary control.

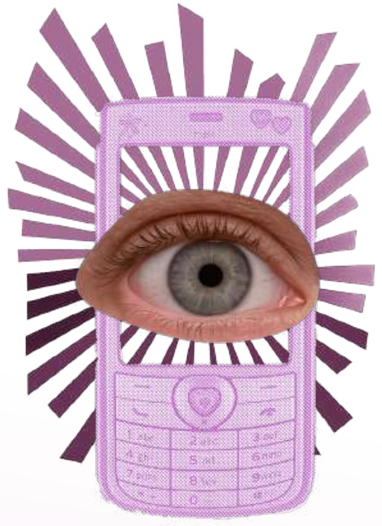
Social Factors

1. **Liquid Modernity (Zygmunt Bauman):**
In a fast-moving and uncertain society, individuals face fragmented lives and a lack of stable identity or emotional belonging.

2. **Emotional Suppression in a High-Pressure Society:**
A large portion of stress-relief toy users are adults experiencing psychological fatigue. The childlike sensory comfort of these toys becomes a substitute mechanism for coping with anxiety and emotional fluctuations.

Research

From Feeling to Feed: How Emotions Become Content



· Emotions have shifted from private experiences to consumable and controllable content, becoming part of the emotion economy.

· More than just stress-relief toys, these visuals have become platform-driven comforts, promoted to attract users and keep them engaged.

· Such content is recommended, imitated, and remixed, serving as a tool to maintain user activity.

· Users transition from passive viewers to co-creators, with emotional expression increasingly managed by the platform.

Sensory Fusion

Synesthesia image from: Mysid - Self-made in Inkscape based on en:Image:Synestheticwiki3.png, Public Domain, https://commons.wikimedia.org/w/index.php?curid=2592788

definition

A phenomenon where one sensory stimulus **triggers another sensory experience**, often used to **break perceptual boundaries** and create **immersive experiences**.

Applications of synesthesia

- Link the squeezing action with sound/color responses
- Simulate natural scenes (e.g. rainy nights, wind, ocean waves)
- Different textures evoke different synesthetic reactions (e.g. soft = safe, transparent = light)
- A healing experience co-constructed by vision, sound, and touch

Interview

I interviewed a variety of squeeze toy enthusiasts and here are representative answers from three different groups:

Ula Zhang University Student

Squeeze toys made of different materials will make me feel differently. The clicking sound of the silicone when pinched under pressure is like pressing the delete key on a keyboard. The translucent material is like jelly, and the resistance of my fingertips sinking in will remind me of slow motion footage.

Jay Zhang Art Practitioner

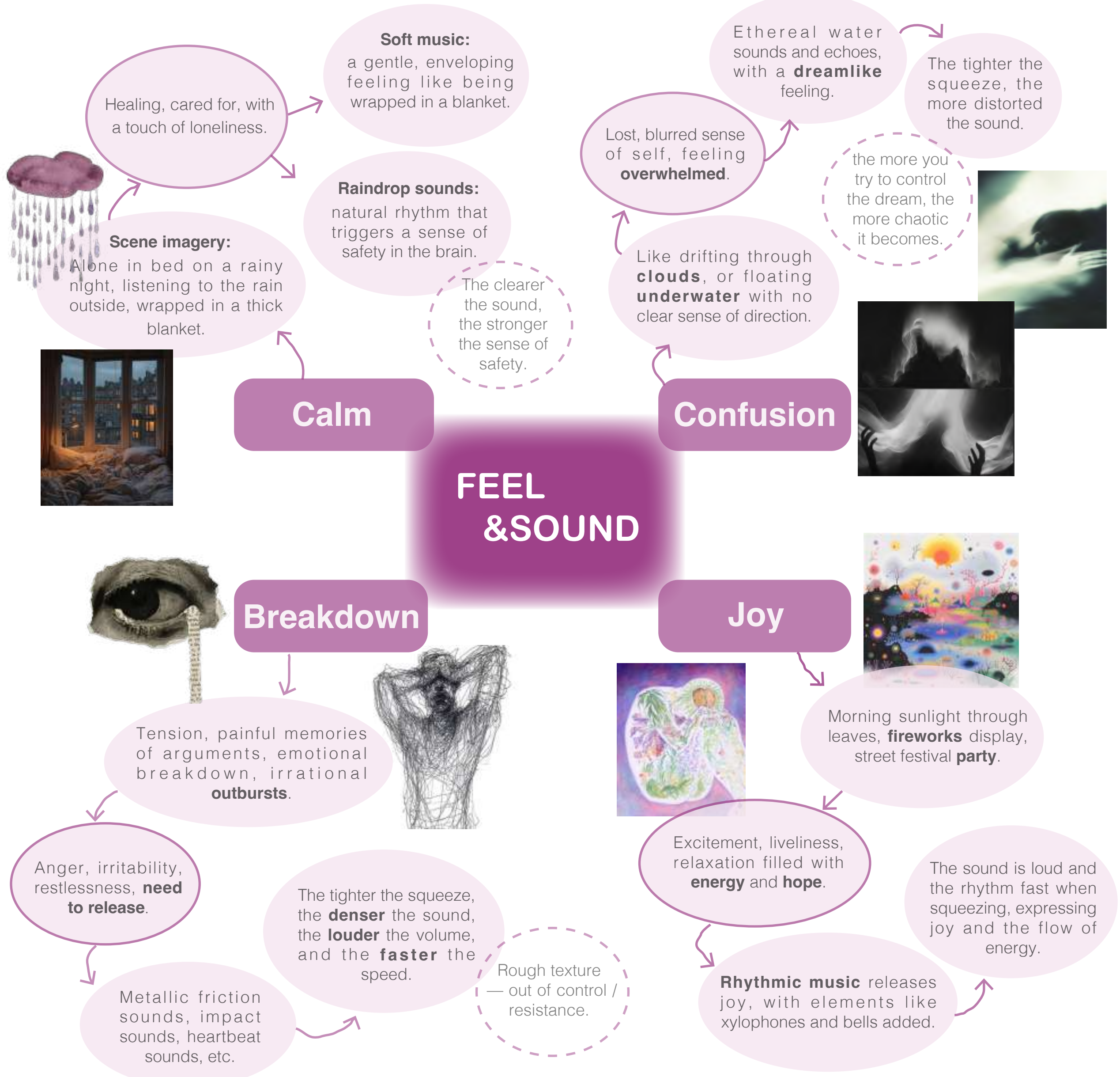
When I squeeze the smooth latex material slow rebound toys, I feel visually surfaced the expansion and contraction of the Morandi colour block.
I've made my own version of a red bean bag with a heat function! The temperature coupled with the sinking feel is like covering a hot water bag, but shaped.

Mr. Li parent of an autistic child

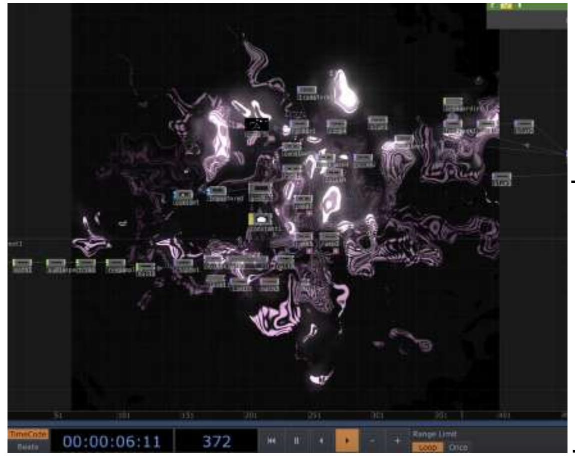
My child often stares at the morphing process of squeeze toys, as if it helps him filter out the clutter of the rest of the environment.
(How do you think digital media can enhance this experience?)
Perhaps an **immersive space** could be created where a child's touch transforms into calming **visuals** and **sounds**, helping them feel safe.

Mind Map

This mind map outlines four interactive emotional scenes, combining tactile feedback, visuals, and sound. Each sound is carefully chosen to reflect and enhance the emotion—calm, confusion, breakdown, or joy—allowing users to feel and express these moods through squeezing gestures.

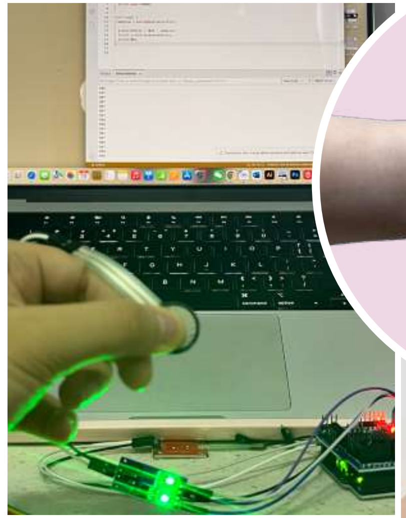


Process



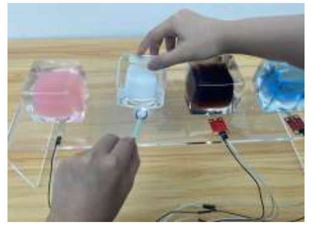
To visualize emotions, I used TouchDesigner to create a flowing light wave effect. Combined with different colors, it resembles the fluctuations of emotional brainwaves, offering an immersive experience.

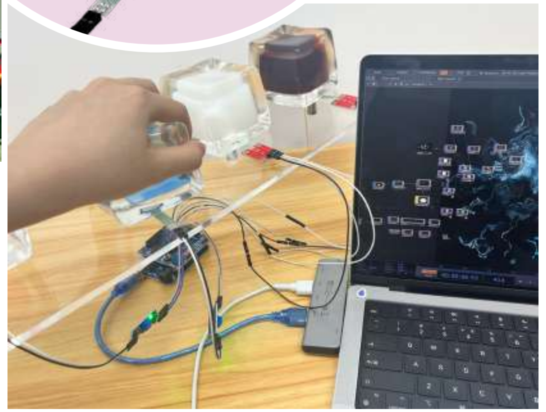
Following the emotional scenarios outlined in the mind map, I chose four squishy objects in representative colors to reflect different emotions.

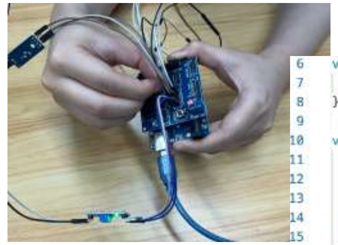


When the hand is placed on it and squeezed firmly, the **pressure sensor** below detects the pressure.

The volume gradually **increases** as pressure is applied. When the hand is **released**, the volume **decreases** slowly, and the visual fluctuations slowly **disappear**.



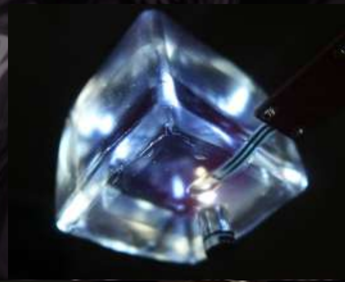


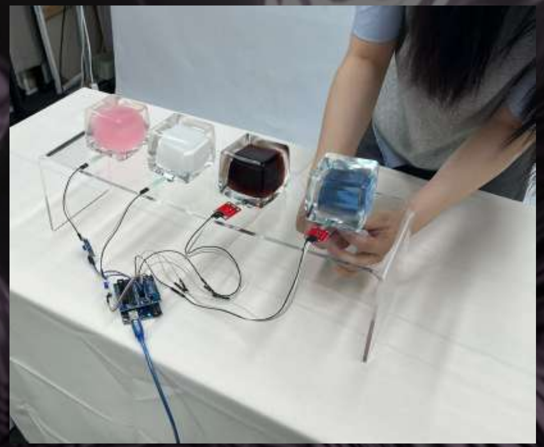


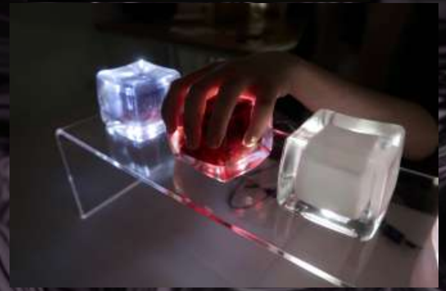
```
void setup() {
  Serial.begin(9600);
}

void loop() {
  rawValue = analogRead(sensorPin);

  pressureValue = 1023 - rawValue;
  Serial.println(pressureValue);
  delay(100);
}
```

I connected four pressure sensors using an Arduino UNO R3 combined with an expansion board for easier integration.

I attached a light to each squishy, making them look like jelly.



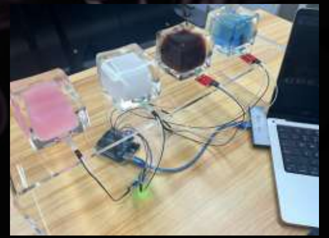




Problem Solving

After testing two thin-film pressure sensors, I found their response to light pressure was sluggish. I added a moving average filter to smooth the data and improve visual flow.

```
void setup() {
  Serial.begin(9600);
  for (int i = 0; i < 10; i++) {
    readings[i] = 0;
  }
}

void loop() {
  total = total - readings[index];
  readings[index] = analogRead(sensorPin);
  total = total + readings[index];
  index = (index + 1) % 10;

  average = total / 10;

  Serial.println(average);
  delay(10);
```



Emotional Fusion

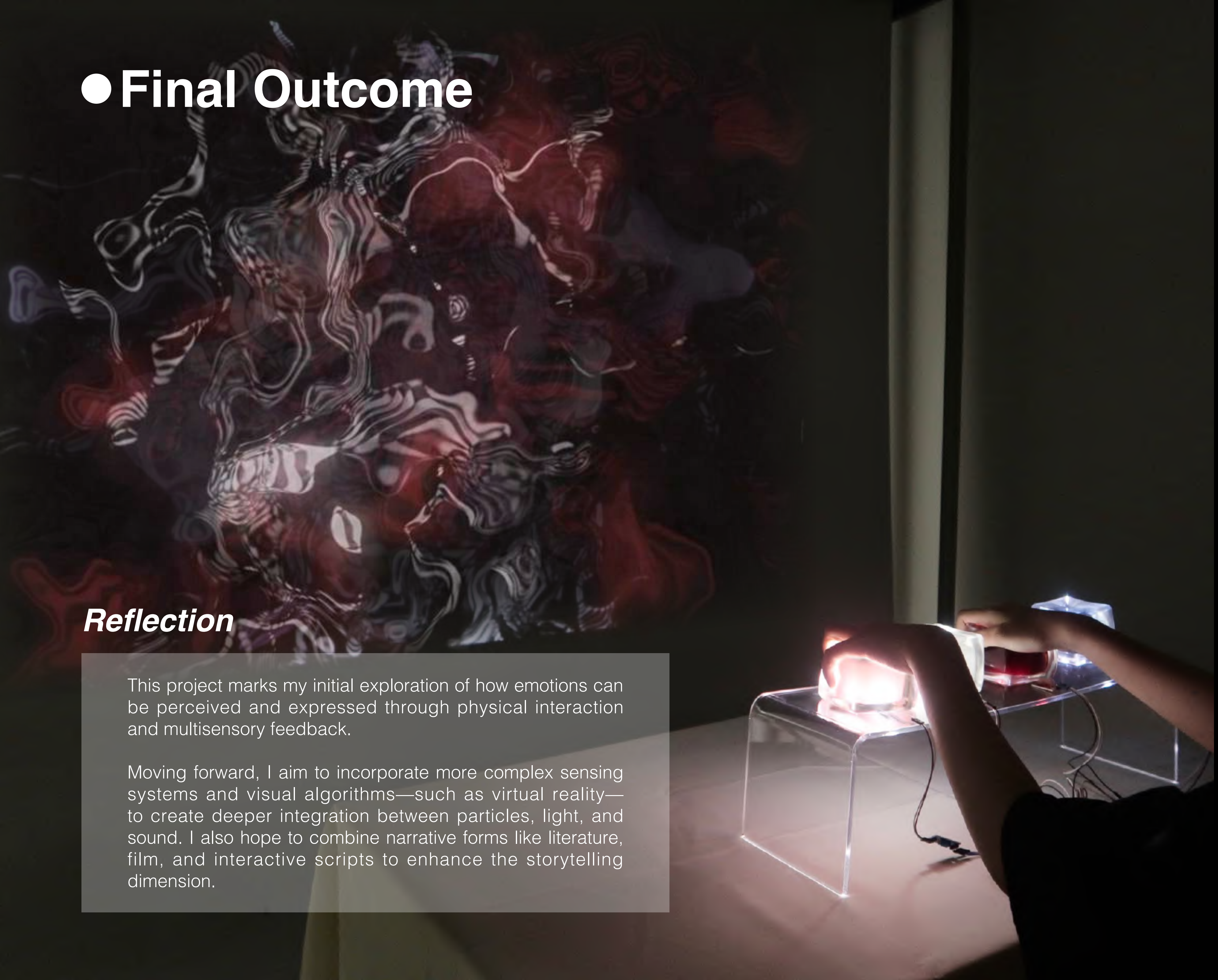
When two or more sensors detect pressure, it triggers a real-time fusion of their respective audio-visual scenes.

This interaction invites participants to explore the complexity of emotions— such as how joy and anxiety, confusion and anger can coexist at the same time.

Participants shared: **"**These combinations made me feel like watching fireworks in the rain, or listening to a lullaby during a thunderstorm.**"**

"The contrast between crashing sounds of breakdown and joyful music even reminded me, strangely, of scenes from horror or thriller films.**"**

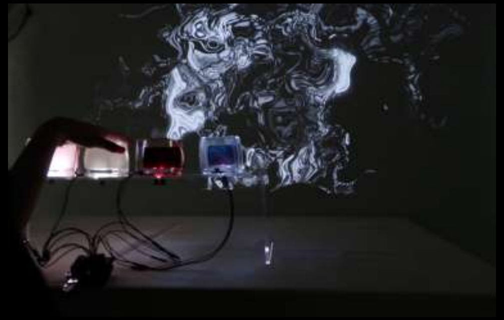
● Final Outcome

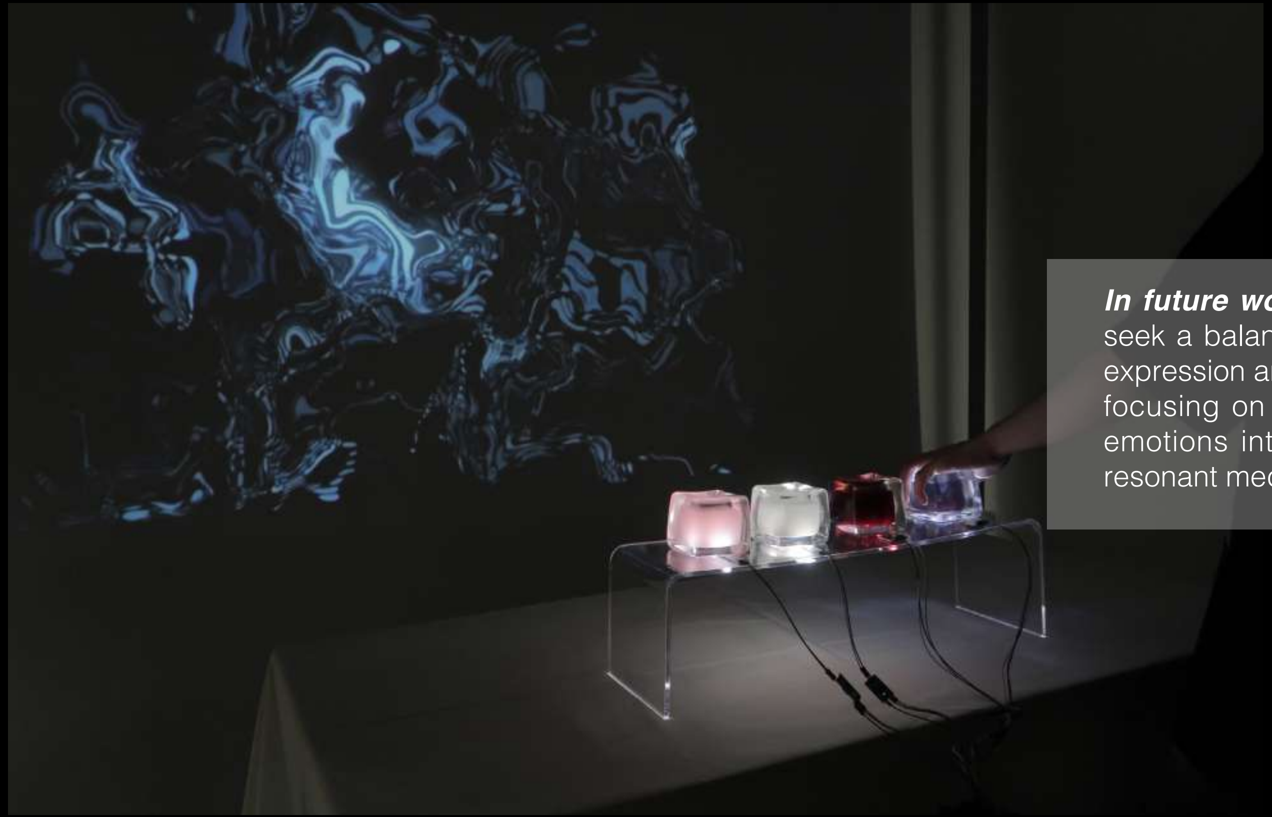


Reflection

This project marks my initial exploration of how emotions can be perceived and expressed through physical interaction and multisensory feedback.

Moving forward, I aim to incorporate more complex sensing systems and visual algorithms—such as virtual reality—to create deeper integration between particles, light, and sound. I also hope to combine narrative forms like literature, film, and interactive scripts to enhance the storytelling dimension.





In future works, I will continue to seek a balance between emotional expression and technical execution, focusing on transforming abstract emotions into interactive, socially resonant media experiences.

Personal Project 03

Tree-Ring Archive

—Mixed-Media Installation

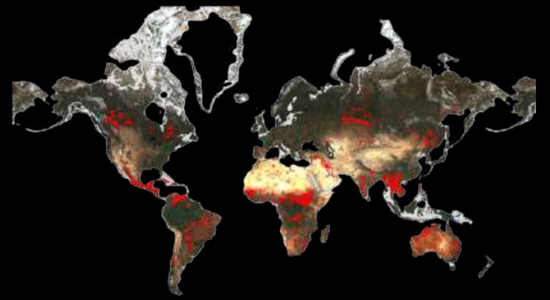
Video Link：*https://youtu.be/odr5w6xug3o*

This project takes forest wildfires as its starting point, exploring how the historical information embedded within tree rings can be transformed into perceivable data.

The device integrates image recognition, sensor interaction, and real-time motor control. As the disc rotates, the system scans the tree cross-section, extracts ring contours, and visualizes growth anomalies caused by environmental changes or disasters. Tree-Ring Record combines speculative design with technical prototyping, offering a new approach to understanding the impacts of disasters and ecological memory.

Background

Inspiration



On January 7, 2025, a wildfire broke out in southern California, USA, covering more than 10,000 acres in total. The fires in this month caused at least 29 deaths, and significant economic losses..

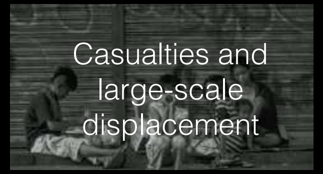
In addition, wildfires have occurred around the world in the past decade.

Weaknesses in Fire Control

In terms of fire investigation, traditional methods (such as power equipment inspection, eyewitness testimony, satellite monitoring, etc.) have been able to find the starting point and approximate cause of the fire. However, there are still many deficiencies, and even in some cases aggravated the impact of disasters.

- Lack of systematic documentation of fire impacts
- Inappropriate forest management strategies
- Lack of real-time monitoring technology
- Slow ecological recovery

Results

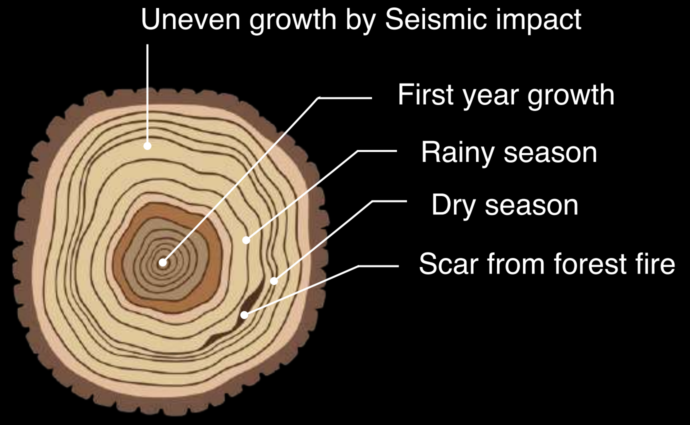


Research

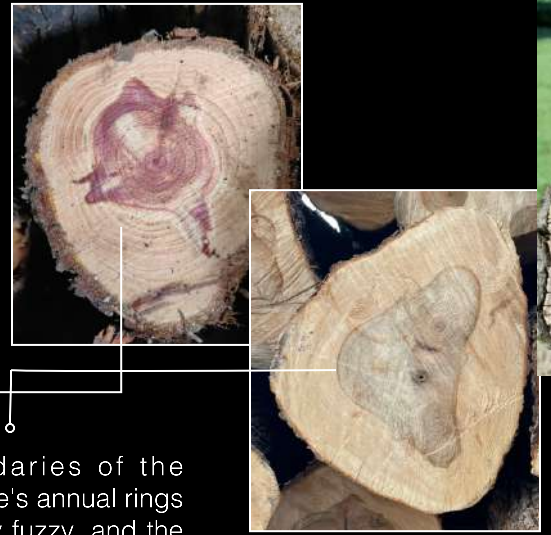
In the Field: Capturing Tree Rings

I conducted field studies of different cross-sections of trees in different settings such as forests and parks.

· Informations from tree rings:



· The clarity of tree rings varies significantly between different tree species:



The rings of the cypress trees have a dark core and a light outer ring. Each layer is clearly defined and easy to identify.

The boundaries of the camphor tree's annual rings are relatively fuzzy, and the contrast between early and late wood is not obvious.

· Abnormal shape of tree rings:

The center of the trunk often decays due to fungal infection, destroying the wood structure.

The rings across the cracks are difficult to track and count, and it is necessary to find corresponding growth rings in adjacent areas to continue tracking. The cracks may also reflect different environmental events.

Characteristics of forest fires

The United Nations Intergovernmental Panel on Climate Change (IPCC) pointed out in its report that climate change is increasing the frequency and intensity of extreme weather events.

The data shows that the area burned by forest fires increased by about 5.4% each year between 2001 and 2023. Forest fires now cause nearly 6 million more hectares of tree cover loss each year compared to 2001, an area roughly the size of Croatia.

About tree rings

How to record environmental changes

The width, density and isotopic composition of tree rings can be used to reconstruct past temperature, precipitation and humidity. Variations in cell size and density determine the clarity of the growth rings.

When a fire occurs, low-intensity surface fires will leave charred scars on tree trunks. The subsequent growth rings of surviving trees will grow callus tissue around the charred scars, forming recognizable traces of the fire history.

How to analyse tree rings

Dendrochemistry

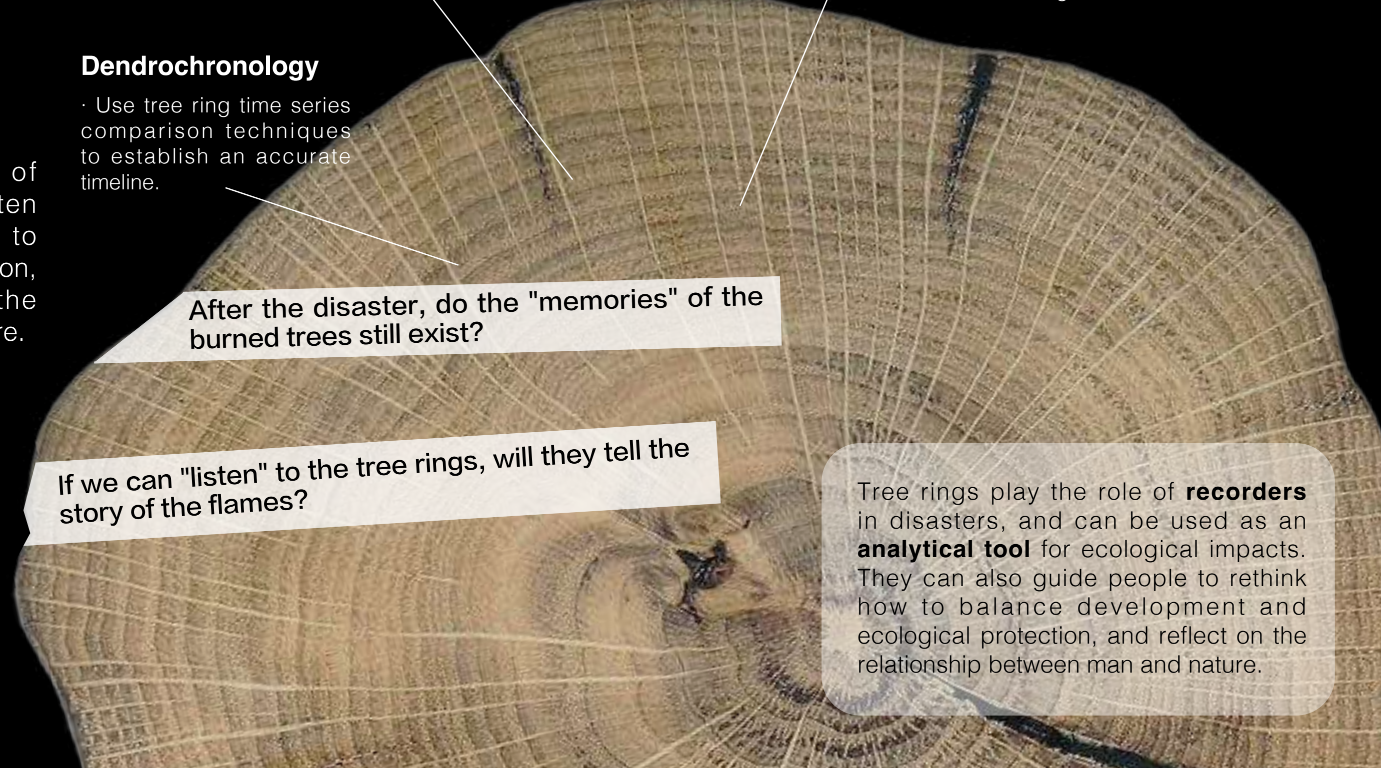
· The chemical analysis of tree rings

Technology-assisted Methods

· X-ray Densitometry
· High-Resolution Microscopy
· Machine Learning & AI

Dendrochronology

· Use tree ring time series comparison techniques to establish an accurate timeline.



Tree rings play the role of **recorders** in disasters, and can be used as an **analytical tool** for ecological impacts. They can also guide people to rethink how to balance development and ecological protection, and reflect on the relationship between man and nature.

For dead trees, cross-sections can be cut for analysis.

For living trees, a tool can be used to penetrate the tree's core and extract a cross-section along its diameter.

The markings in the sample can be used to reconstruct the complete growth rings.

Analytical method

1) Build a "fire ring dataset" : Collect tree cross-sections and train AI to identify fire impacts.

⬇

2) Use convolutional neural networks (CNN) for pattern recognition to improve the accuracy of tree ring classification.

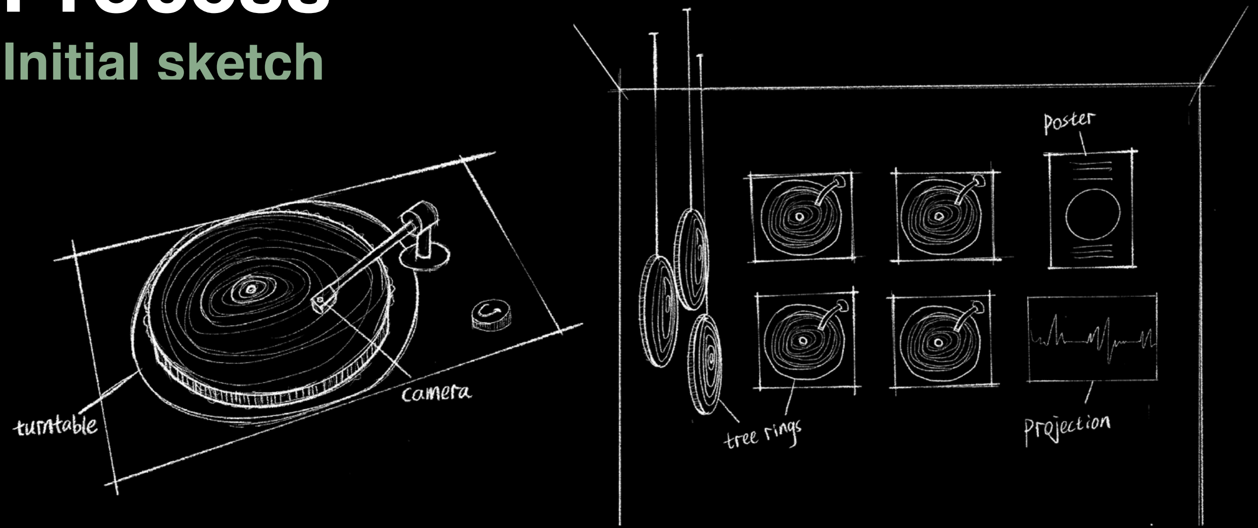
⬇

3) Realize a portable "tree ring scanning" tool: Use the SqueezeNet model to run the tree ring detection model on a mobile device.

Reference：de Geus, A.R. et al. (2020). An analysis of timber sections and deep learning for wood species classification. Multimedia Tools and Applications, 79(45-46), pp.34513–34529. doi:https://doi.org/10.1007/s11042-020-09212-x.

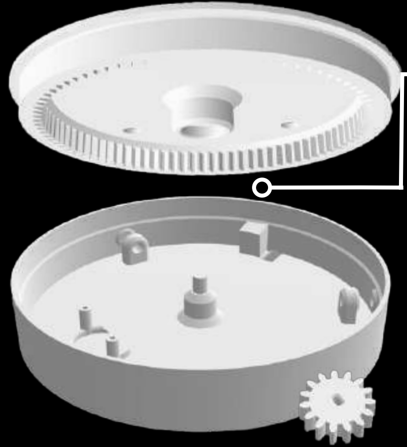
Process

Initial sketch

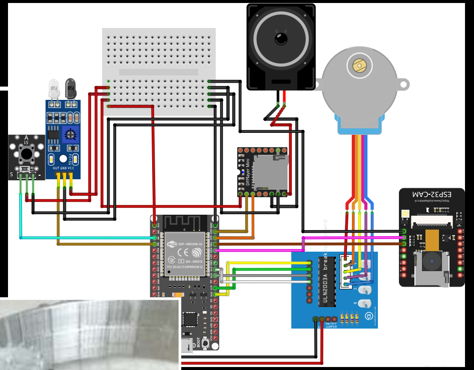


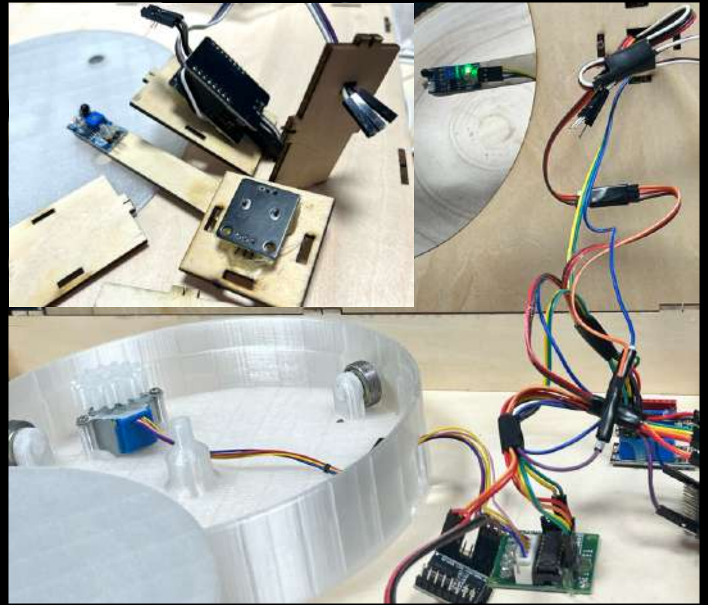
My initial idea was to design an installation modelled on a record player. A cross-section of a tree would act as a record and spin it around the device.
In order to show the installations in action, I initially wanted to hang the installation indoors for the filming and made sketches of the location. But after many attempts it was found that an outdoor setting would be more fitting. So I ended up integrating the installations into the forest.

Device Construction

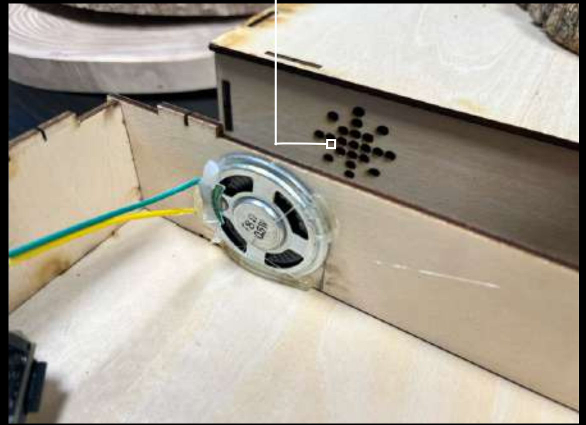


The disc section for housing the internal circuitry was 3D printed. The wiring layout is illustrated as follows.





Music can be played like a regular record player during the recognition process.



System Workflow

[Camera Module]

⬇ ESP32-CAM connects to Wi-Fi and provides a live image stream via /cam-hi.jpg

[Python + OpenCV Live Preview] ←→ [Tree-Ring Detection → Perimeter Extraction]

⬇

[Serial Transmission or Data Logging]

⬇

[Sensor Input Check ← Infrared + Analog Sensors]

⬇ YES

[ESP32 Controls Motor Rotation]

⬇

[Rotating Reader Device + Data Visualization]

Tree Ring Detection

Sensor-Controlled Motor Logic

Potentiometer Rotation Angle → Camera Detects Tree Rings → Start Rotation & Data Recognition

The ESP32 microcontroller checks two conditions:

- Infrared sensor: detects whether a tree ring is in position
- Analog sensor: determines if the threshold is triggered

→ When both are satisfied, the motor activates and the disc begins to rotate.

```
void loop()
{
    if ((analogRead(tu)>3000)&&(digitalRead(ir)==0)){
    stepper.setSpeed(100);
    stepper.runSpeed();
    }else{
stepper.stop();
}
```

```
void handleJpgHi()
{
  if (!esp32cam::Camera.changeResolution(hiRes)) {
    Serial.println("SET-HI-RES FAIL");
  }
  serveJpg();
}
```

OpenCV was used for image processing. After grayscale conversion and binarization to enhance edges, contours were extracted with findContours and their perimeters calculated using arcLength. Closed contours represent complete rings, while open ones indicate partial edges. The results are visualized by drawing the contours.

```
# Convert to grayscale image
img = cv.imread('C:\\Users\\lefthand\\Desktop\\b.jpg', 0)

# Applying Thresholds
ret, thresh = cv.threshold(img, 127, 255, 0)

# Find Contours
contours, hierarchy = cv.findContours(thresh, cv.RETR_TREE, cv.CHAIN_APPROX_SIMPLE)

cnt = contours[0]
perimeter = cv.arcLength(cnt, closed=True)
print(f'Perimeter: {perimeter}')

#ser = serial.Serial(port='COM3', baudrate=9600, timeout=1)
```

Selected rings reflect abnormal growth signals, hinting at possible environmental disturbances.

Growth Data

This is the tree-ring data from a damaged pine tree:

Ring ID (Inner to Outer)	Tree Age	Perimeter (px)	Change from Previous	Notes
R1	1	250	—	Young tree, slow growth
R2	2	365	+115.0	Normal growth
R3	3	480	+115.0	Normal growth
R4	4	620	+140.0	Rapid growth
R5	5	700	+80.0	Stable growth
R6	6	730	+30.0	Slight stagnation
R7	7	615	−115.0	Significant growth reduction (Possible environmental stress: drought or pests)
R8	8	800	+185.0	Growth recovery
R9	9	910	+110.0	Consistently healthy
R10	10	760	−150.0	Sharp ring narrowing (Possible disturbance event)
R11	11	780	+20.0	Slow recovery
R12	12	880	+100.0	Gradual normalization
R13	13	990	+110.0	Full recovery
R14	14	1110	+120.0	Mature phase
R15	15	1180	+70.0	Growth slowdown, reaching stability
R16	16	1195	+15.0	Early aging stage
R17	17	1200	+5.0	Aged, minimal growth
R18	18	970	−230.0	Severe growth loss (Partial damage or incomplete sampling)
R19	19	1005	+35.0	Slow post-disaster recovery
R20	20	1050	+45.0	Steady recovery

Pine

Elm

Locust

Paulownia

I also used the device to explore tree rings from different species and found a limitation: trees with lighter-colored rings, such as Paulownia, are more difficult to recognize accurately.

Outcome

In the final moment, the machine slows, the disc halts, and the tree's story rests in silence.
Through this speculative device, what was once unread and forgotten beneath bark and soot becomes traceable, visible, and almost audible.
Each groove, each gap, each interruption...
This is not merely a tool for precision, but a gesture toward remembrance.

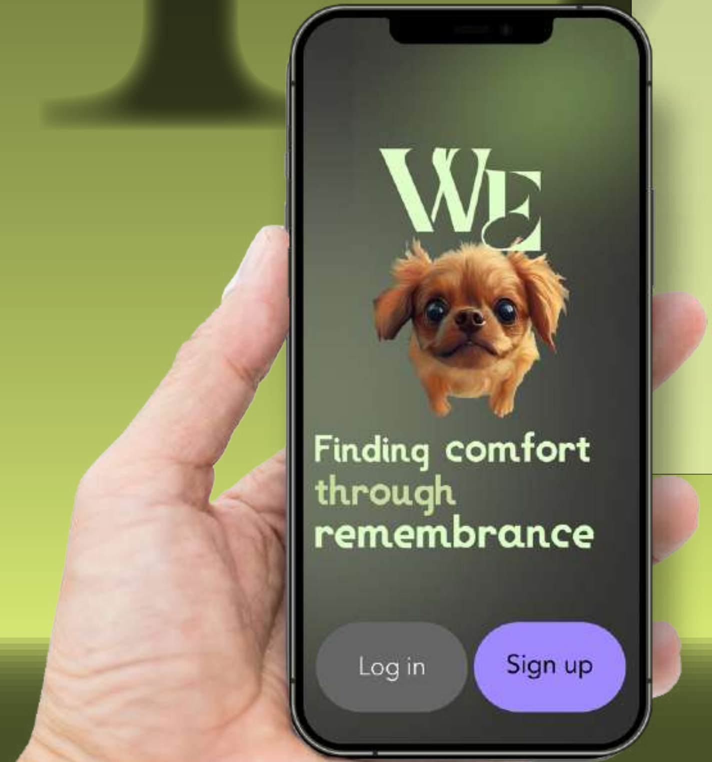
Personal Project 04

Wagging Echoes

—January-February 2025

Emotional Design

Interface Design



INTRODUCTION

There is a wonderful emotional connection between people and pets. The small bodies of pets bring infinite energy and comfort to humans. After experiencing the pain of losing a pet, I have been thinking about how people should face the departure of relatives of different species more calmly? This project may help me and guide people with similar experiences. I hope that they can relieve their emotions in the commemoration experience and continue the connection with their pets.

BACKGROUND

The history of the relationship between pets and humans In Chinese dynasties

Tang (618-907)

Pets in the early Tang Dynasty were more manifestations of instrumentality, such as horses. Many pets are auspicious symbols for them.

Song (960-1279)

During this period, the society is light and martial arts, and pets often appear in poetry written by literati.

Yuan (1271-1368)

The lion is a representative of the royal pet of the Yuan Dynasty. The court used a very unique and particular method to feed them for viewing.

Ming (1368-1644)

After the cats died in the palace of the Ming Dynasty, they even used golden coffin. During this period, the emperor consumed a lot of money on pets.

Qing (1636-1912)

The Qing Dynasty attached great importance to the cultivation of hunting dogs and was equipped with a large number of full-time dog raising personnel.

Modern

Now people pay more attention to the health and happiness of animals. The development of science and technology has improved the quality of raising, and strengthened the interaction between people and pets.

· **Conclusion:** Pets are developed by poultry, and the history of pets in China can be traced back to the primitive hunting period. Since the Tang Dynasty, animal species have gradually enriched, and the style of pet breeding has expanded from the royal class to the home of the people, and pet culture has been further developed.

The impact of pets on human emotions

—Attachment Theory

Pets are often regarded as **important attachment figures**, providing a sense of **security** and **comfort**.

The attachment between people and pets is similar to that between close human beings.

Service pets can help patients **find solace** and help college students **relieve stress**.

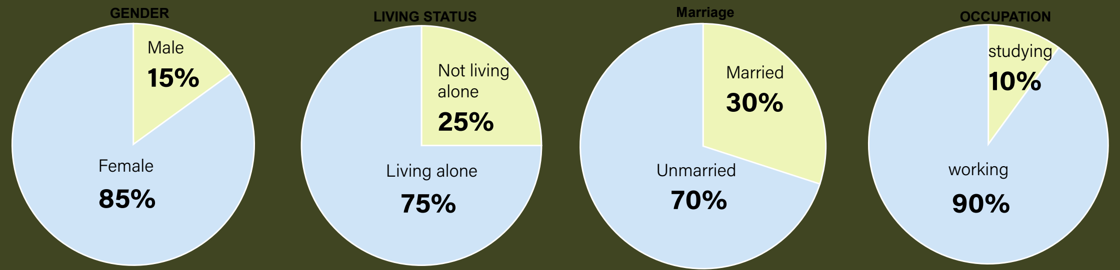
In mental health, pet companionship can **relieve depression and anxiety**.

RESEARCH

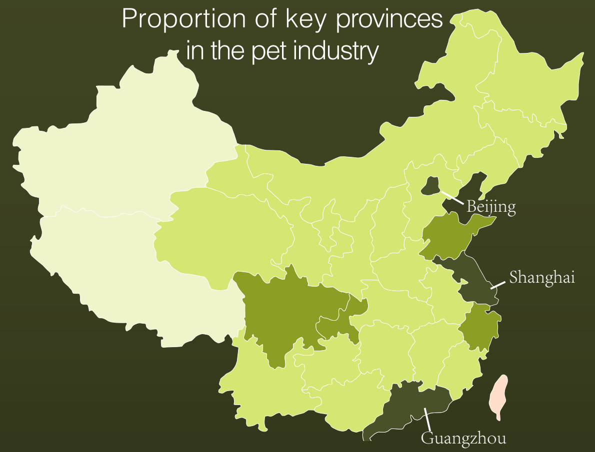
Questionnaire

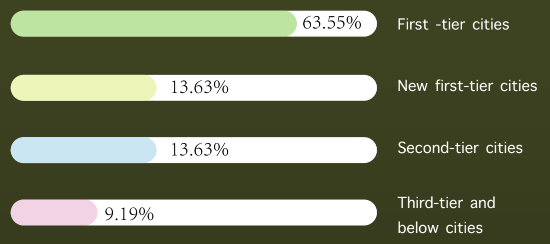
Interviewer Background

A total of 35 people were interviewed, ranging in age from 16 to 48. Almost all of the interviewees' pets were cats or dogs.



Current Status of Pet Funerals in China

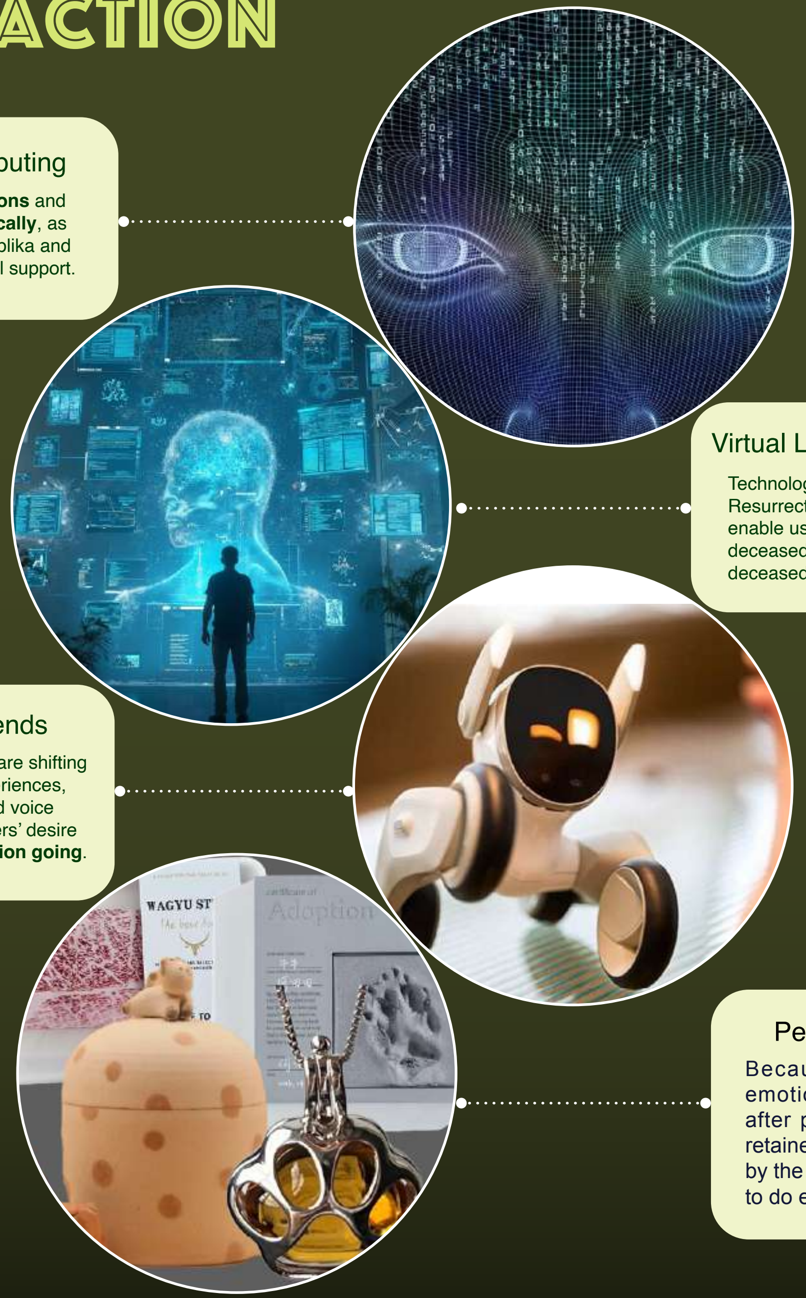




Pet funeral process

Reserve | transport service | Body treatment | Cremation | Ashe

AI AND EMOTIONAL INTERACTION



Affective Computing

AI can **detect emotions** and **respond empathetically**, as seen in apps like Replika and Woebot for emotional support.

Virtual Life & Digital Legacy

Technologies like Microsoft's Digital Resurrection patent and HereAfter AI enable users to **reconnect** with the deceased, allow users to talk with deceased loved ones.

Emerging Trends

Pet memorial services are shifting toward interactive experiences, with **AI-driven** pets and voice interaction meeting users' desire to **keep the conversation going**.

Pet Memorial Items

Because pets have deep emotions between people, after pets, they are usually retained and commemorated by the owner in various forms to do emotional sustenance.

DESIGN STAGE

Brand Aim

Wagging Echoes helps pet owners navigate the grief of losing their beloved pets with heartfelt digital memorials and interactive features, offering a platform of love, memories, and solace.

Logo Sketch

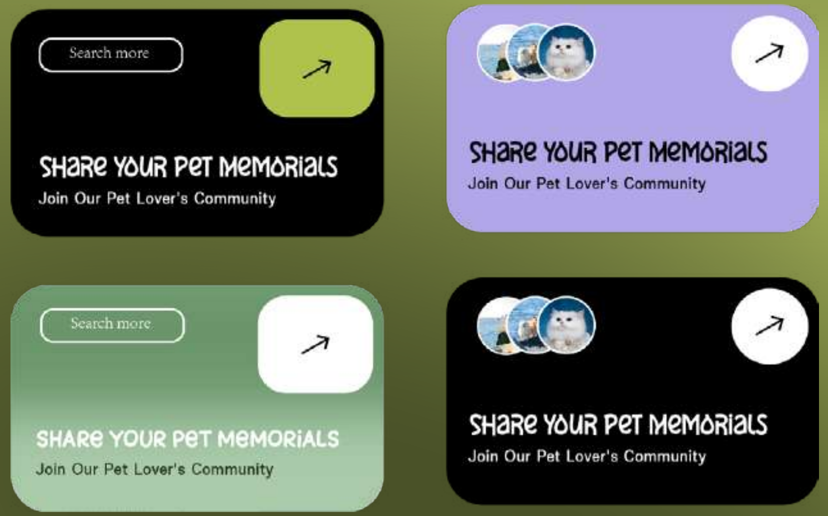
The logo incorporates the initials "WE," symbolizing "us," to reflect the deep bond between pet owners and their pets—a connection that turns individuals into a unified "WE."

Final Logo

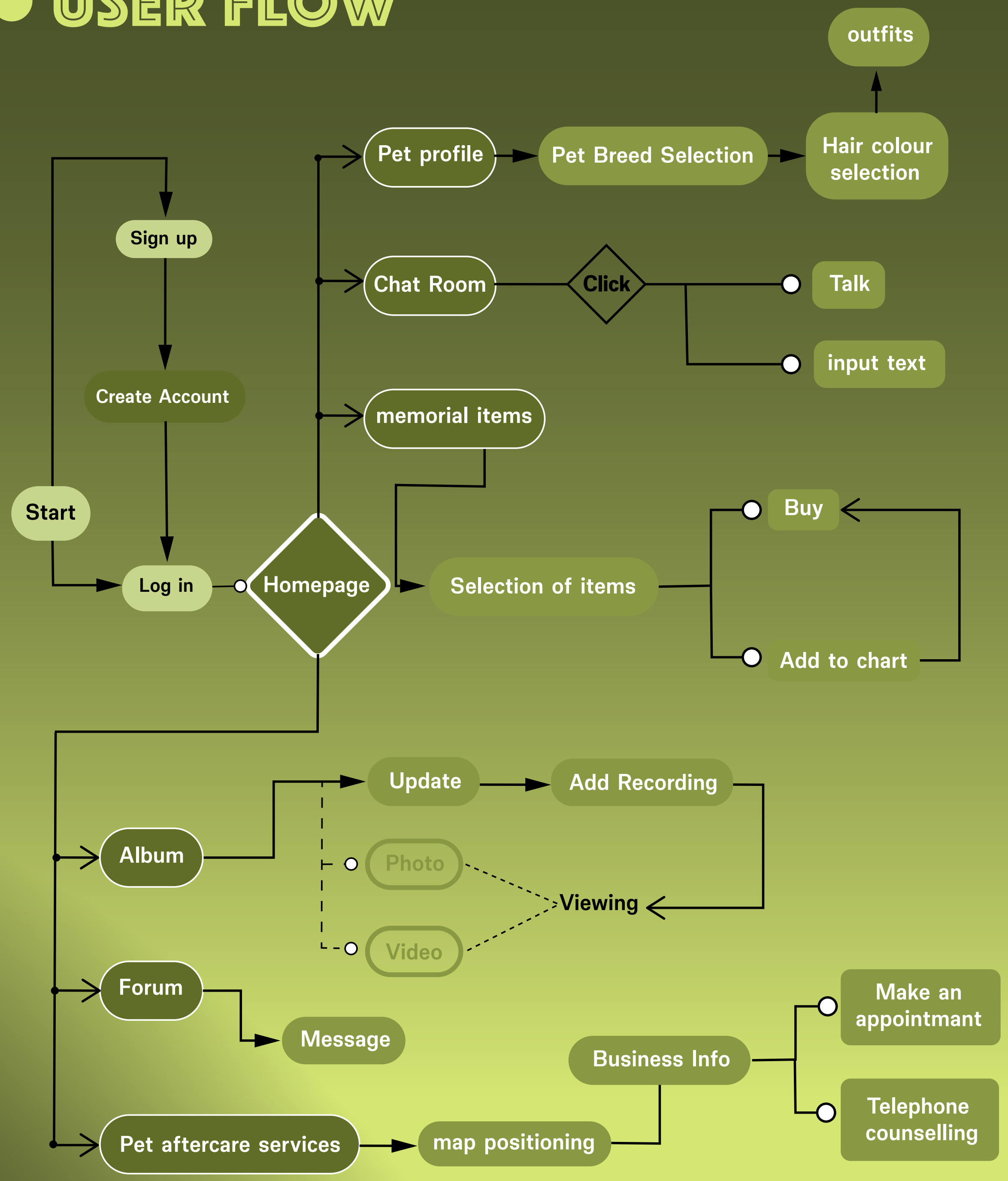
Icon Design

Colour Palette

#cde658	#4d4d44	#a8c680	#a28bff
R 205 G 230 B 88	R 77 G 77 B 68	R 168 G 198 B 128	R 162 G 139 B 255

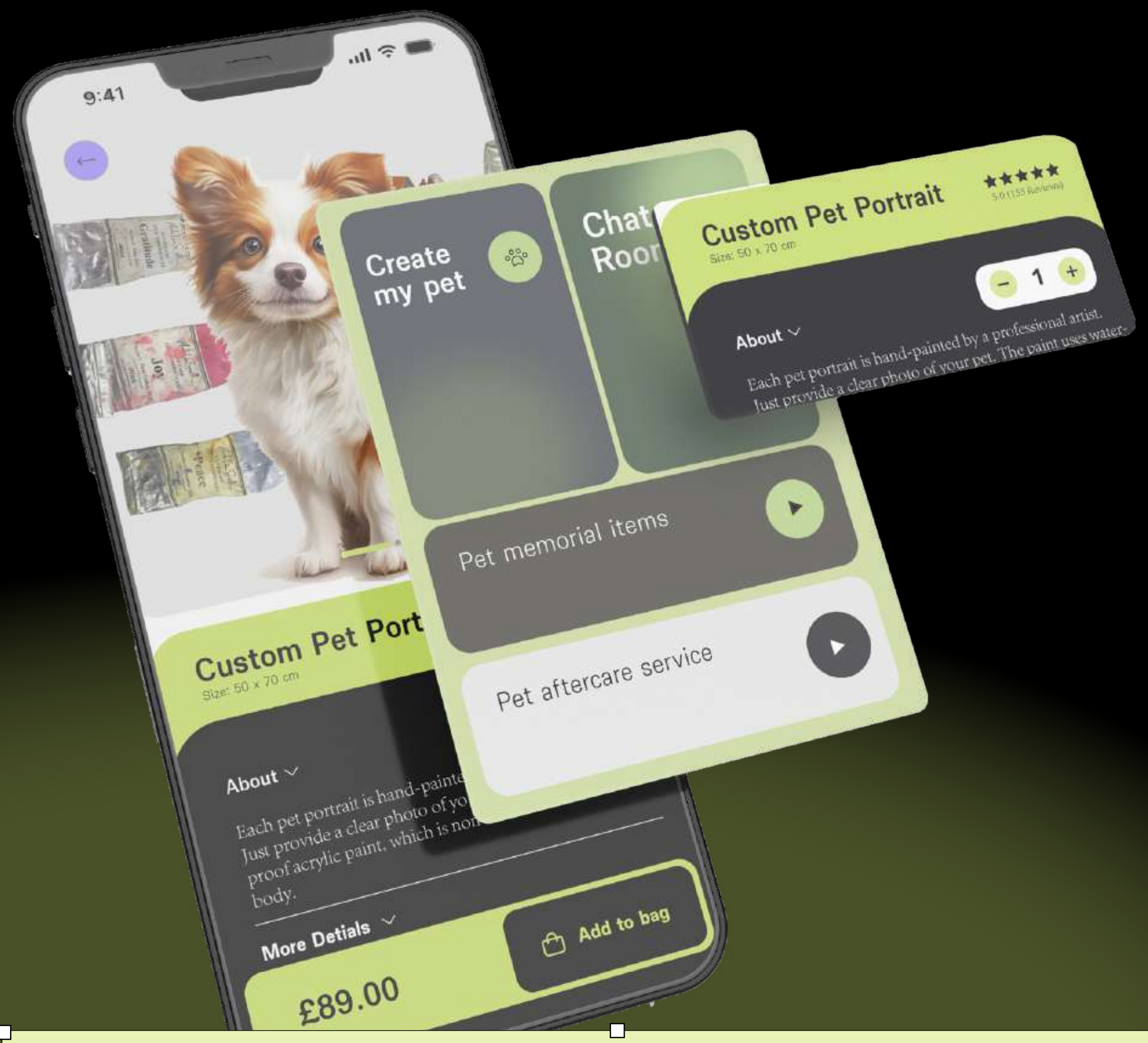


USER FLOW



INTERFACE DESIGN

OUTCOME



Kafka said that the meaning of life lies in the end. Death makes us move forward and motivates us to achieve our goals, learn, love, and create.